Kshitij Duraphe

Boston, MA

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Education

Boston University **Boston, MA**

M.S., Electrical and Computer Engineering (with Thesis) Sep 2022 – May 2024

Advisor: Prof. Joshua Semeter

Degree GPA: 3.8/4.0

Selected coursework: Cosmic Plasma Physics · Fourier Optics · Quantum Mechanics and Semiconductor Physics · Quantum Computing · Signal Processing · Machine Learning · Advanced Algorithms · Advanced Discrete Mathematics

College of Engineering Pune **Pune, India**

B.Tech., Electrical Engineering (Minor: Computer Science) Aug 2018 – June 2022

Advisors: Prof. Archana Thosar, Prof. Suhas Kakade

GPA: 9.11/10.0 (Department Bronze Medalist)

Selected coursework: Probability Theory and Statistical Inference · Optics and Modern Physics · Electromagnetism · Numerical Methods · Vector Calculus · Partial Differential Equations · Ordinary Differential Equations · Complex Analysis

Research Interests

High-Energy Astrophysics; Foundation Models; Mechanistic Interpretability; Space Physics; Time-Domain Astrophysics.

Publications

1. **State-Dependent X-ray Variability in Cygnus X-1: A 12-Year NuSTAR Timing Study of Accretion Flow Geometry**
   K. Duraphe, G. Bhatta, K. Mandar, C. Khanal, et al.
   The Astrophysical Journal, 2026
2. **Multi-Epoch NuSTAR Spectral Analysis of Cygnus X-1: Coronal and Reflection Properties Across Spectral States**
   K. Duraphe, G. Bhatta, A. A. Zdziarski, et al.
   Submitted to The Astrophysical Journal, 2026
3. **Cross-Modal Attention is Overparameterized: The Sufficiency of Modality-Level Alignment**
   K. Duraphe, K. Mohamed, N. Morgan
   Submitted to NeurIPS Main Track, 2026

4. **Smartphone Carrier Phase TEC: A Study Across Ionospheric Spatio-Temporal Scales**
   N. Servan-Schreiber, J. Semeter, ***K. Duraphe***, et al.
   Space Weather, 2026
5. **The Platonic Universe: Do Foundation Models See the Same Sky?**
   K. Duraphe, M. J. Smith, J. F. Wu, S. Sourav *(co-first author)*
   NeurIPS ML4PS Workshop, 2025 — Spotlight (top 1%) expanded follow-up submitted to NeurIPS 2026 main track
6. **Optimizing Solar Panel Tilt using Machine Learning Techniques**
   K. Duraphe, S. Kakade, et al.
   GPECOM, 2021

Theses

1. **Data-Driven Techniques to Advance Our Understanding of the STEVE Phenomenon**
   Kshitij Duraphe, Joshua Semeter
   M.S. Thesis, Department of Electrical and Computer Engineering, Boston University, 2024
2. **Design of an Automated Radio Telescope for Observing the 21 cm Hydrogen Line**
   Kshitij Duraphe, Archana Thosar
   B.Tech. Thesis, Department of Electrical Engineering, College of Engineering Pune, 2022

Research Experience

University of Zielona Góra — **Zielona Góra, Poland (Remote)**

Research Assistant — Mar 2025 – Present

Advisor: Prof. Gopal Bhatta

- First-authored a 12-year NuSTAR archival timing study of Cygnus X-1 (*The Astrophysical Journal*, 2026), mapping accretion-flow geometry across spectral states via hard X-ray spectral-timing analysis.
- Identified a previously unrecognized failed state transition and characterized state-dependent variability (power spectra, rms-flux relations, lag-frequency spectra), placing direct constraints on the corona–disk geometry near the black hole.
- Reduced and analyzed the complete public NuSTAR Cyg X-1 archive with HEASoft / NuSTARDAS / Stingray / AstroPy; led manuscript writing and referee response.
- Currently leading a follow-up multi-epoch **spectral** study of Cyg X-1 with Andrzej A. Zdziarski (NCAC Warsaw), submitted to *The Astrophysical Journal*, characterizing coronal and reflection properties across spectral states and investigating the potential failed state transition.

UniverseTBD Collaboration — **Remote**

Researcher — Jul 2024 – Present

Collaborators: Dr. Michael J. Smith, Dr. John F. Wu

- Co-first author on *The Platonic Universe* (NeurIPS ML4PS 2025 **Spotlight**, top 1% expanded follow-up submitted to NeurIPS 2026 main track with M. J. Smith and J. F. Wu). Designed the evaluation framework comparing foundation-model representations across SDSS, DESI, JWST, and other surveys under model and data scaling.

- Creator of the open-source platonic-universe package (Python / PyTorch): data loaders, evaluation harness, reproducible configs, test infrastructure.
- Invited AstroAI talk at the Harvard Center for Astrophysics (Jan 2026) and NeurIPS ML4PS oral presentation (Dec 2025). Spotlight award (top 1% paper) NeurIPS ML4PS 2025.
- Used NCSA DeltaAI compute allocation **PHY250286** — *The Platonic Universe: Do Foundation Models See the Same Sky?* — 6,000 GPU-hours (PI: M. J. Smith; Sep 2025 – Sep 2026); actively running training and evaluation experiments under the allocation.

Space Physics Lab, Boston University **Boston, MA**

Graduate Research Assistant Oct 2022 – May 2024
Advisor: Prof. Joshua Semeter

MS Thesis: Data-Driven Techniques for STEVE Jul 2023 – May 2024

- Characterized STEVE morphology and kinetics from high-resolution citizen-science imagery; developed computer-vision tracking methods that quantified distinct westward velocities for columnar (13 km/s) and picket-fence (1 km/s) features.
- Developed and contributed to various open-source Python and C++ libraries for high-cadence GNSS carrier-phase processing and Total Electron Content (TEC) analysis; identified a candidate two-stage TEC signature of STEVE passage distinct from typical substorms.
- Developed automated STEVE detection for noisy All-Sky Imager data by adapting faint-source astronomical algorithms and evaluating deep-learning models (ConvNeXt), with modular superresolution, inpainting, and time-series classification components.

NASA Life on Mars Initiative - BU Center for Computing and Data Sciences Jul 2023 – Jan 2024

- Developed GPS-signal propagation models and spatiotemporal interpolation of sparse Martian-ionosphere TEC data (classical and deep-learning baselines) from MARSIS.

MS Project — Plasma Dynamics of STEVE Oct 2022 – May 2023

- Ran 3D forward-modeling simulations of the ionospheric plasma regime conducive to STEVE formation using the open-source GEMINI3D fluid-electrodynamic model; analyzed density structures, temperature enhancements, and flow shears driven by STEVE-like conditions.

IIT Bombay **Mumbai, India**

Research Intern — Bayesian Binary-Star Analysis May 2021 – Aug 2021

- Performed Bayesian analysis of the eclipsing binary QX Cas using PHOEBE and MCMC; developed a custom 3-body dynamics simulator with a differential-evolution solver to constrain parameters of a hypothesized tertiary companion.

College of Engineering Pune **Pune, India**

Undergraduate Research Assistant Jan 2021 – May 2022
Advisors: Prof. Archana Thosar, Prof. Suhas Kakade

Signal Processing Lab — Data-Driven Solar Panel Tilt Optimization Jan 2021 – Apr 2021

Advisor: Prof. Suhas Kakade

- Formulated optimal solar-panel tilt as a regression problem over multi-year irradiance and meteorological time-series; benchmarked gradient-boosted, kernel-based, and shallow-network estimators against the standard latitude-based heuristic.
- Showed measurable energy-yield gains across heterogeneous Indian and American climatological zones; first-authored the resulting publication at IEEE GPECOM 2021.

B.Tech. Thesis: 21 cm Radio Telescope Aug 2021 – May 2022

- Designed and built a high-gain (20 dB) pyramidal horn radio telescope for 21 cm hydrogen-line observations (Ansys HFSS-simulated waveguide/feed).
- Integrated RTL-SDR + LNA front-end with **Python and C++** WOLA-FFT spectral-analysis software (VIRGO spectrometer-based); recovered a partial galactic rotation curve from 21 cm emission.

Naxxatra Equinox Initiative **Bangalore, India**

Research Intern — Stellar Structure and Compact Objects Jul 2020 – Oct 2020

- Derived the equations of stellar structure incorporating hydrostatic equilibrium and relativistic electron-degeneracy pressure; numerically solved the coupled ODEs with a 4^{th}-order Runge-Kutta scheme in Python to compute the white-dwarf mass-radius relation and determine the Chandrasekhar limit.

Selected Talks and Posters

- *The Platonic Universe: Do Foundation Models See the Same Sky?* (invited, long version) AstroAI, Harvard CfA, Jan 2026
- *The Platonic Universe: Do Foundation Models See the Same Sky?* (oral) NeurIPS ML4PS, Dec 2025
- *Using high-rate dual-frequency cellphones to study the April 8^{th} solar eclipse* CEDAR Workshop, Jun 2024
- *Optimizing Solar Panel Tilt using Machine Learning Techniques* GPECOM, Dec 2021

Awards and Honors

- NeurIPS ML4PS 2025 — **Spotlight Paper** (top 1%) 2025
- MS Ambassador, Boston University ECE Department 2024
- 2^{nd} / 100+ international teams, MIT iQuHack 2023 (IBM x Covalent Challenge) 2023
- Bronze Medalist, Electrical Engineering Department, College of Engineering Pune 2022

Teaching, Mentoring, and Open-Source Community

Computational Research Access NEtwork (CRANE) Physics **Winter 2024, 2025**

TA / Mentor

- Topics: signal processing, Python, numerical methods, machine learning, PIC simulations. Wrote tutorial materials and supported students during hands-on sessions across both cohorts.

ENG EC520 Image Processing (Graduate Course), Boston University **Spring 2023**

Grader
Instructor: Prof. Janusz Konrad

- Graded problem sets and projects on filtering, compression, reconstruction, and computer-vision algorithms for a graduate-level image-processing course.

ArkOS — MIT Student Information Processing Board (SIPB) **Cambridge, MA**

Contributor — DevOps and Documentation 2025 – Present

- Lead DevOps and documentation for an open-source local-LLM agent platform; maintain GitHub Actions CI/CD, contributor onboarding, and developer-facing documentation.

COEP Astronomy Club **Pune, India**

Head of Projects 2018 – 2022

- Mentored 30 undergraduates on processing astronomical data, operating telescopes, and building reflector optics.
- Ran outreach sessions at schools for autistic children on identifying constellations in the night sky.

Key Skills

Astronomy tooling: AstroPy, HEASoft, NuSTARDAS, Stingray, PHOEBE, GEMINI3D, FITS / HDF5 I/O

Scientific Python ecosystem: NumPy, SciPy, Matplotlib, pandas, PyTorch, scikit-learn

Languages: Python (primary), C, C++, SQL, MATLAB

Statistical and numerical methods: Bayesian inference (MCMC), spectral-timing analysis, time-series analysis, ODE/PDE numerical solvers, Runge–Kutta

Software engineering: Git / GitHub, GitHub Actions CI/CD, pytest, Sphinx-style documentation, open-source contribution and code review

Instrumentation: Ansys HFSS, radio receiver chains (RTL-SDR + LNA), antenna design

Industry Research Experience

Thespian Labs **Somerville, MA**

AI Engineer Nov 2025 – Feb 2026

- Developed state-of-the-art **Text2Motion** foundation models using **VQ-VAE** motion tokenization; ran multimodal pre-training on 5000+ hours of human-performance time-series and DARTControl-style **reinforcement-learning post-training** for controllable motion synthesis.
- Released the open-source MLOps library jobber for programmatic research-job submission to cloud platforms.

Absentia Technologies **Boston, MA**

Founding Machine Learning Engineer Jan 2025 – Nov 2025

- Developed state-of-the-art **vision-language models (VLMs)** for multimodal video understanding; combined large-scale pre-training with **supervised fine-tuning and alignment post-training** on domain-specific data.
- Implemented distributed training pipelines with **PyTorch FSDP** (multi-GPU, mixed-precision) to train foundation models beyond single-device memory, handling sharding, checkpointing, and fault recovery.

Halo AI (Columbia-incubated stealth) **New York, NY (Remote)**

Founding AI Engineer Dec 2023 – Aug 2024

- Researched on-device **federated** and **ensemble** LLMs for agentic assistants under tight compute and memory constraints, including ablations on federated-aggregation strategies.
- Systematic study of **model-compression** techniques — pruning, knowledge distillation, and 8-bit post-training quantization — and their trade-offs for latency, memory footprint, and task accuracy (50% size reduction, 80% inference speedup, 90% accuracy retention).

Spatialise **Noordwijk, The Netherlands (Remote)**

Geospatial Machine Learning Engineer Feb 2025 – May 2025

- Developed a **multimodal spatiotemporal foundation model** over multispectral satellite imagery for soil-health remote sensing; combined deep learning with statistical priors (Gaussian-process regression, clustering) to produce calibrated predictions.

The KeelWorks Foundation **Oak Harbor, WA (Remote)**

Software Engineer, ML Applications Jul 2024 – Jan 2025

- Synthetic-data generation research via **Mistral-7B** fine-tuning (back-translation, sequential style representation); agentic LLM systems with RAG pipelines for large-scale document retrieval and question answering.

Astronomy Schools and Workshops

Inter-University Center for Astronomy and Astrophysics (IUCAA) **Pune, India**

Introductory Summer School in Astronomy and Astrophysics May – Jun 2020

Review Experience

- United States Research Software Engineer Conference (US-RSE) 2025, 2026